TrueRehab RF Poster Speaking Script

AEII 2026 | English | Prepared 4 November 2026

Use this script beside the submitted poster. Italic cues guide pointing and are not spoken. Keep the slide deck as a backup.

Introduction

[Point to the project title]

Hello, and welcome to our project, TrueRehab-RF.

We are developing a contactless system to help monitor movement during rehabilitation. Our goal is to study whether millimeter-wave radar and AI can help identify changes in movement quality and fatigue-related compensation.

The idea is to look at how an exercise is performed, as well as whether the person can complete it.

Background and problem

[Point to Background & Problem]

Our project focuses on a simple question: when someone completes an exercise, are they performing the intended movement, or are they using another part of their body to compensate?

For example, during a biceps curl, a person might lift their shoulder or move their trunk while continuing to raise the weight.

The weight may still reach the same height, even though the movement has changed. This is why repetition count alone cannot describe the whole exercise.

We want to study whether radar can help identify these changes across several repetitions.

Innovation concept

[Point to Innovation Concept]

Our proposed approach uses radar to observe movement through reflected radio signals. The sensor transmits a signal and receives reflections from the person and surrounding objects.

Processing these reflections gives us information about distance and movement. We can then investigate how the observations change during an exercise.

The long-term goal is to monitor exercise without requiring a camera or sensors attached to the person during normal use.

For our first experiment, we are focusing on one exercise: the biceps curl.

Starting with one exercise lets us keep the setup more consistent before exploring other rehabilitation movements.

Operation flowchart

[Point to Operation Flowchart]

This flowchart shows our proposed processing pipeline.

The radar collects reflections, and signal processing organizes the measurements into useful representations. The planned AI stage would then study patterns across a sequence of repetitions.

We are interested in changes in movement speed and consistency, and whether these relate to independently observed compensation.

There is one hardware update to explain. The submitted poster shows our original AWR1843BOOST and DCA1000 design. Because of hardware availability, our current prototype uses the TI AWR2944PEVM, with onboard processing and USB communication.

AI development and validation

[Point to AI analysis and Key development]

To develop the AI, we first need real recordings with meaningful reference labels.

We have tested our MyoWare muscle sensor, which provides a muscle-activity reference. We plan to combine synchronized recordings with movement annotations to study relationships between the radar signal and the exercise.

Muscle activity alone does not establish fatigue, so we still need to define and validate our fatigue labels.

We would evaluate the model using independent participants or sessions. This helps us check whether it works beyond the recordings used for training.

Dashboard and live demonstration

[Point to Dashboard, then show the live demo]

At this stage, we can receive real radar measurements and display point clouds and heatmaps.

A point cloud shows the estimated positions of detected reflections. Heatmaps help us inspect reflected signal strength across distance and direction, or distance and speed.

These displays help us check the measurements before using them for model development.

The dashboard illustrates the feedback we aim to provide. Its fatigue and movement-quality scores are demonstration outputs. We have not yet validated those scores using real rehabilitation data.

Expected benefits

[Point to Distinctive point and the intended users]

Our intended users include people undergoing rehabilitation, caregivers, and physical therapists.

If the approach proves reliable, a future system could summarize movement changes and help a therapist review an exercise session.

The final design aims to provide contactless monitoring during home exercise. Reference sensors such as MyoWare support development and validation, rather than being a requirement for everyday use.

Next steps and closing

[Point to Key development, then return to the project title]

Our next milestone is to collect synchronized biceps-curl recordings and establish which radar features can reliably describe the movement changes that matter.

That evidence will guide model development and later testing on the Raspberry Pi prototype.

Ultimately, we hope this research can support useful feedback during home exercise and help physical therapists understand a person's progress.

Thank you. We would be happy to show you our prototype and answer your questions.